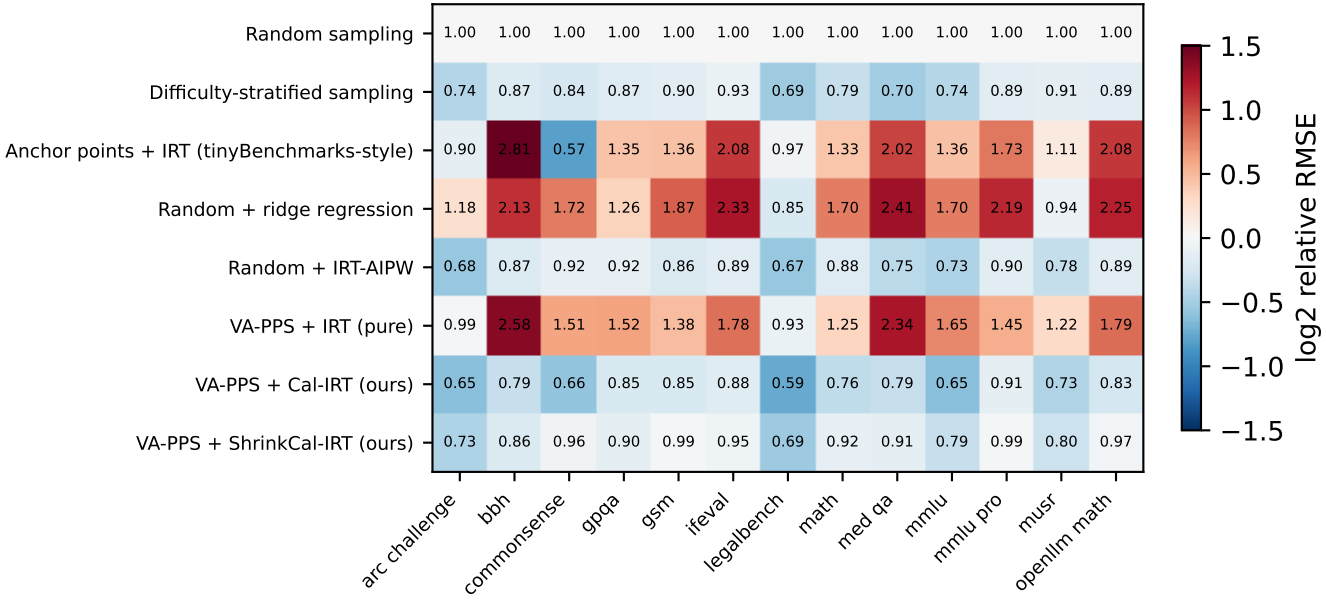
Random sampling
Difficulty-stratified sampling
Anchor points + IRT (tinyBenchmarks-style)
Random + ridge regression
Random + IRT-AIPW
VA-PPS + IRT (pure)
VA-PPS + Cal-IRT (ours)
VA-PPS + ShrinkCal-IRT (ours)
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.74 0.87 0.84 0.87 0.90 0.93 0.69 0.79 0.70 0.74 0.89 0.91 0.89
0.90 2.81 0.57 1.35 1.36 2.08 0.97 1.33 2.02 1.36 1.73 1.11 2.08
1.18 2.13 1.72 1.26 1.87 2.33 0.85 1.70 2.41 1.70 2.19 0.94 2.25
0.68 0.87 0.92 0.92 0.86 0.89 0.67 0.88 0.75 0.73 0.90 0.78 0.89
0.99 2.58 1.51 1.52 1.38 1.78 0.93 1.25 2.34 1.65 1.45 1.22 1.79
0.65 0.79 0.66 0.85 0.85 0.88 0.59 0.76 0.79 0.65 0.91 0.73 0.83
0.73 0.86 0.96 0.90 0.99 0.95 0.69 0.92 0.91 0.79 0.99 0.80 0.97
arc challenge
bbh
commonsense
gpqa
gsm
ifeval
legalbench
math
med qa
mmlu
mmlu pro
musr
openllm math
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
log2 relative RMSE